De La Salle Holy Cross College

NOVEMBER EXAMINATION

2023

Grade 10

LIFE SCIENCE

Paper 2

Examiner: Mrs G. Tolley

Moderators: Mrs D. Webb, Ms B. Hannweg

Time: 60 minutes

Marks: 40

Name: _____ **Class_____**

PLEASE READ THROUGH THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This paper consists of 8 pages. Please check that your question paper is complete.
2. You are required to write an *argumentative essay* addressing the question:
3. It is in your interest to write legibly and to present your work neatly.

Question One - Essay	Possible mark	Actual mark
	40	

Remember that God is with you and will help and inspire you in all that you do

The following statement 'sets the scene' for the topic of the argumentative essay.

> Previously, a former American President, Donald Trump, made comments saying that climate change and global warming due to human activities are foolish and concern over these environmental matters are a waste of time. He went as far as withdrawing the United States of America from the Paris Agreement on June 1st 2017. The central aim of the Paris agreement being to strength the global response to the threat of climate change by keeping a global temperature rise this century well below 2°C.
>
> A study of evolution shows that changes in the environment, such as climate change, very often acts as an evolutionary selection pressure which results in the mass extinction of many organisms that are unable to adapt to the new environmental conditions.

Adapted from Gr 10 November Examination Essay 2017

The climate change that is occurring in the present age will result in the mass extinction of many species.

Using the source material provided as well as your own knowledge, discuss your opinion on the above statement in the form of a 2,5 to 3 page *argumentative essay.*

To answer the question, you are expected to:

- Read the source material carefully and present a debated argument to illustrate your point of view.
- Select relevant information from source A to I.
- It is important to integrate your own relevant biological knowledge.
- Take a definite stand on the question and arrange the information to best develop your argument.
- Write in a way that is scientifically appropriate and communicates your point of view clearly.
- Provide a clear plan of your essay before you start writing.
  NOTE the plan will be marked as part of the assessment of this question.

40 marks

SOURCE A

How is climate change related to mass extinction?

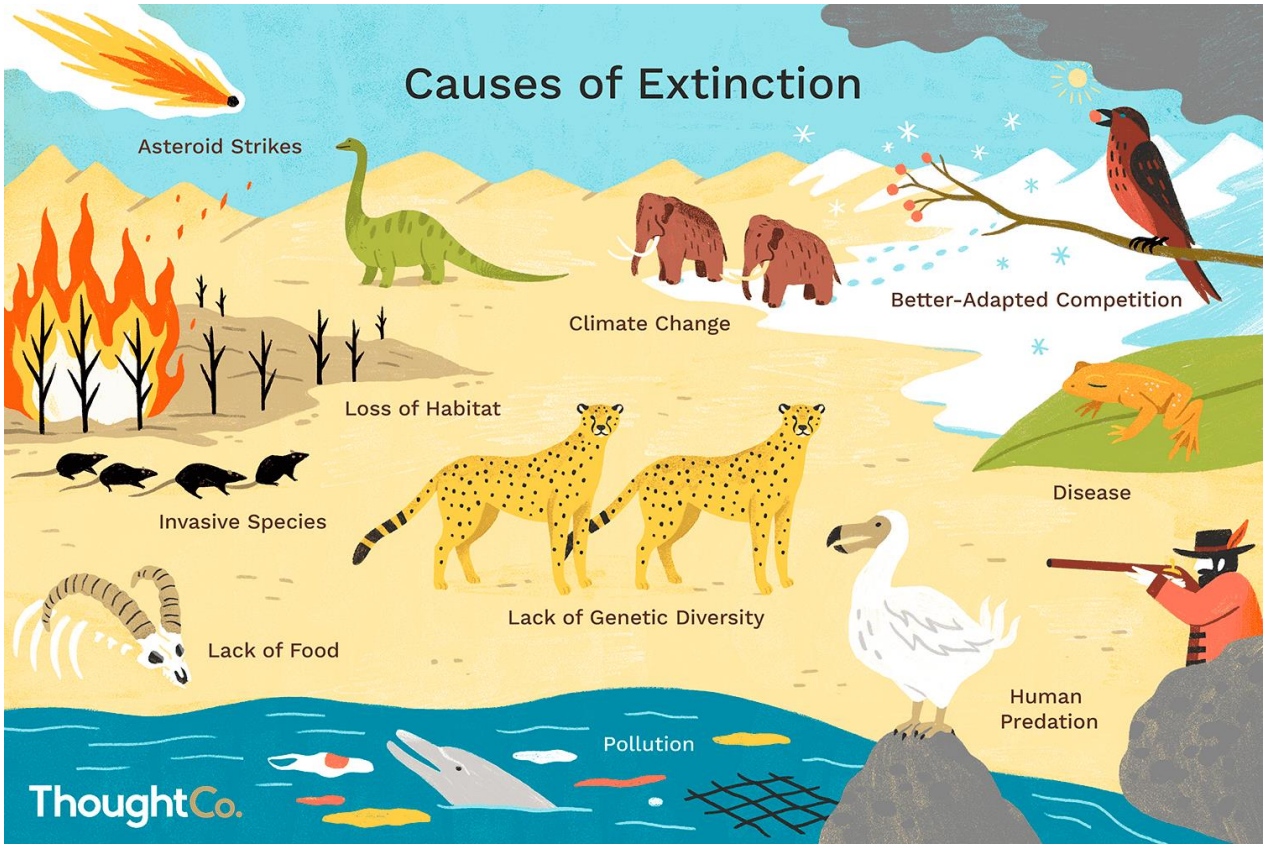
Adapted from: <https://www.thoughtco.com> 2016
(all content rigorously researched and written by experts)

The planet is warming to a degree beyond what many species can handle, altering or eliminating habitat, reducing food sources, causing drought and other species–harming severe weather events, and even directly killing species that simply cannot stand the heat.

SOURCE B

Causes of Mass Extinctions

https://www.thoughtco.com/reasons-animals-go-extinct-3889931
(all content rigorously researched and written by experts)



SOURCE C

By **Heather Scoville**
Science Expert M.A., Technological Teaching and Learning, Ashford University B.A., Biochemistry and Molecular Biology, Cornell University
Updated 18 February 2019

Two current controversial topics is the idea that humans are causing global warming which in turn is leading to the mass extinction of many species if they are not able to evolve to tolerate the increased temperatures.

Global climate change due to the build-up of greenhouse gases, is based on the annual increase in the average global temperature. This increase in temperature seems to be causing many potential environmental issues including the melting of the polar ice caps, more extreme natural disasters like hurricanes and tornadoes, and larger areas are becoming affected by droughts.

Without some greenhouse gases, it would be too cold for life to survive on Earth. However, too many greenhouse gases can have extreme effects on the life that is present.

For example, polar bears that in many ways have become the symbol of climate change, are currently on the endangered species list due to global climate change. Polar bears live in areas where there is very thick ice in the northern polar regions of Earth. They have very thick coats of fur and layers upon layers of fat to keep warm. They rely on fish that live under the ice as a primary food source and have become skilled ice fishermen to survive. Unfortunately, with the melting polar ice caps, the polar bears are finding their once favourable adaptations to be obsolete, and they are not adapting quickly enough. The temperatures are increasing in those areas which make the extra fur and fat on the polar bears more of a problem than a favourable adaptation. Also, the thick ice that was once there to walk on is too thin to hold the weight of the polar bears any longer. Therefore, swimming has become a very necessary skill for polar bears to have.



If the current increase in temperature keeps up or accelerates, there will be no more polar bears. Those who have the genes to be great swimmers will live a bit longer than those who do not possess that gene, but, eventually, all will most likely disappear since evolution takes many generations and there just is not enough time.

There are many other species all over the Earth that are in the same sort of predicaments as the polar bears. Plants are having to adapt to differing amounts of rainfall than what is usual in their areas, other animals need to adjust to changing temperatures, and still, others must deal with their habitats disappearing or changing due to human interference. There is no doubt that global climate change is causing problems and increasing the need for a quicker pace of evolution in order to avoid mass extinctions all over the world.

SOURCE D

Advantages of Global Warming? It's a Bit of a Stretch

Adapted from thoughtco.com/advantages-and-disadvantages-of-global-warming-1434937
(all content rigorously researched and written by experts)

The so-called advantages of climate are out there—if you're really looking but do they compensate for the disruption and destruction wrought by the disadvantages? Again, the answer is no but for die-hard fans of the global warming trend, advantages might include the following suspect scenarios:

- The Arctic, Antarctic, Siberia, and other frozen regions of the earth might experience more plant growth and milder climates.
- Previously uninhabitable areas could become suitable for human habitation and economic development.
- The next ice age could possibly be prevented.
- The Northwest Passage through the formerly icy Canadian Arctic Archipelago could arguably open up to transportation.
- Fewer deaths or injuries to plants, animals and humans would occur due to less severe arctic conditions.
- Longer growing seasons could mean increased agricultural production to feed the expanding human global population could be possible in some areas.
- Previously untapped oil and gas reserves might become available.

SOURCE E

What are the animals most affected by climate change?

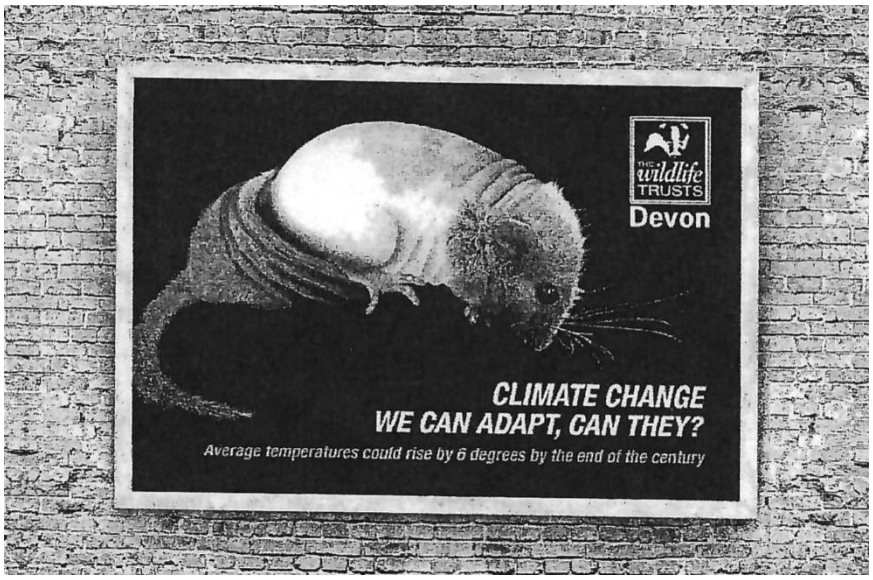
https://www.worldwildlife.org › magazine › animals-affected by Climate Change



Animals Affected by Climate Change

- POLAR BEAR.
- SNOW LEOPARD.
- GIANT PANDA.
- TIGER.
- MONARCH BUTTERFLY.
- GREEN SEA TURTLE.

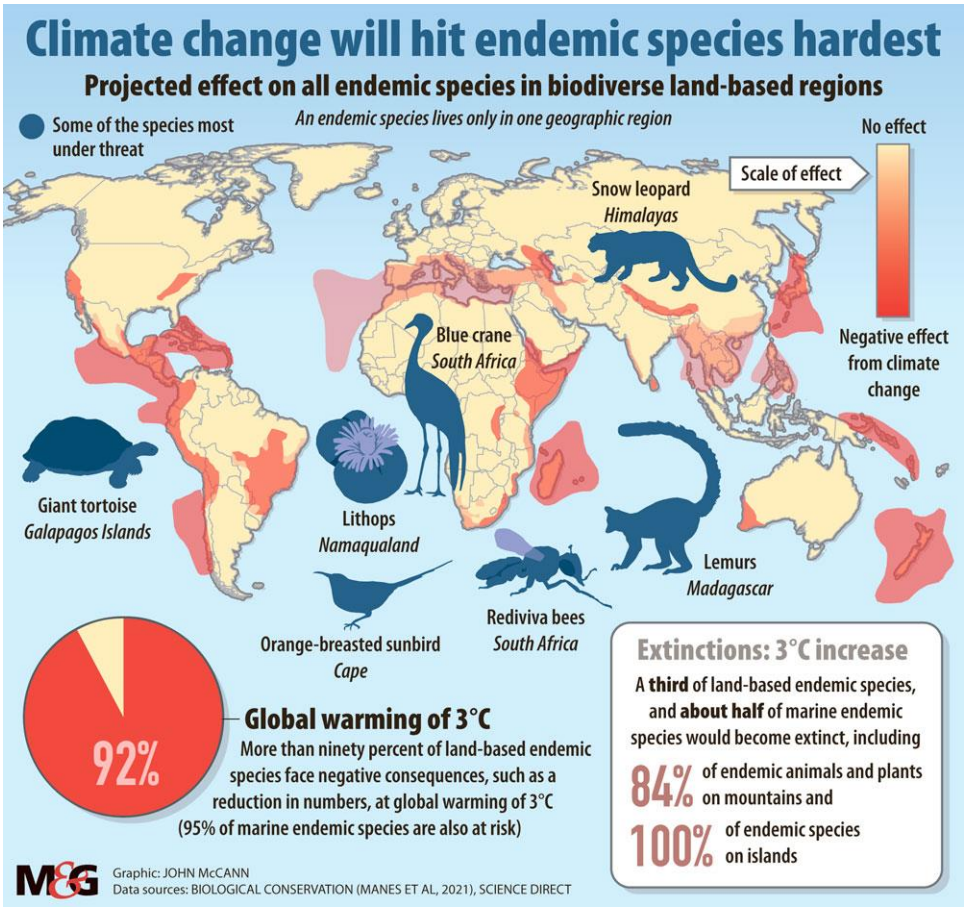
SOURCE F



From:http://www.toucandesign.co.za

SOURCE G

https://mail&guardian.co.za/the-green-guardian/2021-04-17-climate-change-threatens-survival-of-endemic-species-the-most



SOURCE H

WILL CLIMATE CHANGE AFFECT MASS EXINCTION OF A SPECIES?

Adapted from: https://www.quora.com

Asked on Quora: A place to share knowledge and better understand the World; a website where the public can ask experts to answer questions.

Answer from William Hamleck: Science teacher at Stamford High School; M.A. Biology, Western Connecticut State University.

Climate may change too rapidly for adaptation. Survival in a rapidly changing climate will be:

- Acclimatisation
- Migration
- Dispersal
- Ecological niche adaptation

Those species with more generalised niches, wider range of tolerance and greater abilities to disperse, will likely be more successful.

SOURCE I

https://www.graphicnews.com/en/pages/20899/climate-change-highway-to-extinctionarn more about Interior's work on climate change at www.doi.gov/climate.

